

SMART INDIA HACKATHON 2026

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AI-Powered Continuous Cyber Risk Quantification & Investment Optimization Platform

- **Problem Statement ID & Title:** SIH26105 — AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform
- **Theme & Category:** Blockchain & Cybersecurity | Software Solution
- **Ministry / Target Organization:** Ministry of Education's Innovation Cell (MIC) & AICTE (Cyber Security Cell)
- **Team Name & Official ID:** ByteForce_1 (Official SIH Team ID: 180219)
- **Institute Name:** Skyline Institute of Engineering and Technology, Greater Noida
- **Team Leader & Members:** Aryan Baghel (Team Leader), Kirti, Hitesh Chauhan, Avinav Jha, Sharim Khan, Aman Sekh
- **Executive Tagline:** "Bridging the gap between raw cybersecurity telemetry and business decision-making through actuarial rupee loss quantification, Knapsack budget optimization, and tamper-evident blockchain auditability."

Slide 02 | Proposed Solution & Core Innovation (USP)

THE PROBLEM & INDUSTRY PAIN POINTS

✗ Qualitative Jargon Disconnect:

Tools output 'High/Med/Low' labels. Boards and CFOs cannot understand abstract CVE severity or correlate it to balance-sheet liabilities.

✗ Stale Annual Audits:

Traditional risk registers rely on annual manual questionnaires, missing real-time port exposures and dynamic threat actor shifts.

✗ Blind Security Spending:

Enterprises spend millions on random security tools without quantifiable Return on Security Investment (ROSI).

✗ Mutable Audit Logs:

Traditional database logs are vulnerable to insider modification, violating CERT-In 180-day unalterable log mandates.

PROPOSED SOLUTION & INNOVATION (USP)

✓ 1. Actuarial Rupee Quantification (FAIR):

Translates real-time listening socket counts and compute telemetry into prototype Expected Annual Loss (EAL) in ₹ INR.

✓ 2. Pareto Knapsack Budget Optimizer:

Takes explicit institutional budget (e.g. ₹1 Crore) and selects the highest-ROSI defense portfolio delivering optimal risk reduction.

✓ 3. Tamper-Evident Blockchain Ledger:

Anchors risk evaluations, budget plans, and quarantine logs in SHA-256 Merkle blocks, providing cryptographic proof of non-alteration.

✓ 4. What-If Scenario Modeling:

Enables executives to simulate proactive defenses (e.g. closing exposed ports vs patch delays) and view projected exposure reductions.

✓ 5. Zero-Trust Physical Auto-Lock:

Local YOLOv8 monitors operator presence, automatically locking Windows if the operator departs.

Slide 03 | Technical Approach, Methodology & Architecture

LAYER 1: PRESENTATION & SOC

- **Frontend Engine:** React 19 + Vite 8 supercharged dashboard with 11 unified modules.
- **Executive Board View:** Live Enterprise Risk Score, ₹ Exposure, and budget optimization curves.
- **Technical SOC View:** Granular port analyzer, process kill triggers, and printable CISO Audit PDF.
- **Command Palette:** Global quick-action bar (Ctrl+K) for rapid operational response.

LAYER 2: COMPUTATIONAL CORE

- **FastAPI Async Kernel:** High-throughput asynchronous Python micro-services on 0.0.0.0:8000.
- **FAIR Exposure Engine:** $EAL = Threat\ Frequency \times Vulnerability \times Loss\ Magnitude$ in ₹ INR.
- **Knapsack Optimizer:** Evaluates defense control packages to maximize ROSI under budget.
- **What-If Simulator:** Interactive sliders for parameter tuning & MITRE wargame simulations.
- **Credential Scanner:** Regex scanner detecting unmasked Aadhaar, PAN, and leaked API tokens.

LAYER 3: TRUST & AI LAYER

- **SHA-256 Merkle Ledger:** Cryptographically chained blocks with Merkle roots & Proof-of-Work.
- **Tamper-Evident Audit:** Instant verification detecting any un-mined modifications to past logs.
- **SQLite 3 (WAL Mode):** Crash-resilient Write-Ahead Logging local storage hub.
- **YOLOv8 Operator Presence:** Ultralytics YOLOv8-nano tracking operator presence & triggering auto-lock.
- **MediaPipe Touchless Mouse:** Google MediaPipe 0.10 tracking 21 3D hand coordinates for gesture control.

Slide 04 | Feasibility, Viability & Risk Analysis

TECHNICAL FEASIBILITY & VIABILITY

✓ **Commodity Hardware Feasibility:**

Designed as a lightweight local-first architecture running on standard Intel/AMD quad-core CPUs without mandatory cloud GPU servers.

✓ **Zero SaaS Recurring Costs:**

Operates without continuous external API dependencies, ideal for budget-constrained academic and public sector institutions.

✓ **Current Prototype Readiness:**

The full MVP is operational today across all 11 modules with live host telemetry, investment optimization, and blockchain verification.

✓ **Air-Gapped Operational Safety:**

Can be deployed inside closed institutional campus LANs with zero external data telemetry leakage.

POTENTIAL RISKS & MITIGATION STRATEGY

🛡️ **Risk 1: Host Overhead from Telemetry Polling**

Mitigation: Implemented asynchronous non-blocking psutil polling with 3-second throttle, consuming <1% CPU overhead.

🛡️ **Risk 2: Model Accuracy & Subjectivity**

Mitigation: Strict alignment with Open FAIR standards; all formulas, baseline rates, and weights are transparent and configurable by the CISO.

🛡️ **Risk 3: Vision Occlusion / False Auto-Locks**

Mitigation: Configurable 5-second countdown grace period and active keyboard/gesture fail-safe preventing accidental lockouts.

🛡️ **Risk 4: Enterprise Scalability**

Mitigation: Architecture designed with standard REST/Syslog ingress hooks to accept forwarders (Fluentbit, Wazuh) in future iterations.

Slide 05 | Social, Economic & Institutional Impact

TARGET BENEFICIARIES



AICTE & Technical Higher Education:

Helps engineering colleges and polytechnics assess campus cybersecurity posture and allocate grant funds efficiently.



Enterprise CISOs & IT Directors:

Provides mathematically sound business justification for security investments to present before CFOs and Boards.



BFSI & Regulated Sectors:

Assists compliance officers in meeting strict continuous cyber risk quantification and reporting requirements.



State Cyber Police & Forensic Auditors:

Leverages immutable blockchain logs for indisputable post-breach digital forensic evidence.

QUANTIFIABLE ECONOMIC & OPERATIONAL VALUE



Optimized Security Spend:

Eliminates wasteful security tool purchases by prioritizing candidate solutions yielding peak Return on Security Investment (ROSI).



Minimizes Costly Downtime:

Correlates critical open ports with operational downtime, triggering preemptive process mitigation before breach escalation.



Accelerates Audit Readiness:

Reduces periodic compliance audit preparation time from weeks of manual spreadsheet tracking to single-click PDF export.



Non-Repudiation for Regulators:

Provides mathematical certainty to CERT-In and RBI/SEBI auditors that historical logs have not been tampered with.

Slide 06 | Standards, Regulatory Alignment & References

REGULATORY FRAMEWORK CONTROL MAPPING

CERT-In Directions (April 2022):

Satisfies the mandatory 180-day unalterable log preservation directive via SHA-256 Merkle blockchain anchoring.

RBI Cyber Security Framework:

Supports tiered risk threshold alerting and technical resilience disclosures for institutional financial operations.

SEBI Cyber Resilience Framework:

Addresses board-level risk posture visibility, technical asset inventory tracking, and systemic risk governance.

DPDP Act 2023 Compliance Support:

Integrated sensitive credential scanner detects and flags unmasked Aadhaar, PAN numbers, and private keys in scanned paths.

STANDARDS, CITATIONS & 5-MIN DEMO FLOW

The Open Group FAIR Standard (ANSI/FAIR):

The foundational standard for quantitative cyber risk economics and loss magnitude formulation.

NIST Cybersecurity Framework 2.0 & ISO 27001:

Aligns control telemetry across Identify, Protect, Detect, Respond, and Recover functions.

MITRE ATT&CK Matrix v15:

Maps simulated wargame vectors (T1110 Brute Force, T1499 DDoS, T1046 Port Scan).

Live Evaluator Demo Sequence (5 Mins):

Module 01 (Host Status) → Module 02 (Attack Surface) → Module 04 (Rupee EAL) → Module 05 (What-If Simulation) → Module 06 (Knapsack Optimizer) → Module 10 (Live Blockchain Tamper Showdown) → Executive PDF Export.